

```
1 package ques1InventoryManagementSystem;
2 public class Product {
3     int productId;
4     String productName;
5     int quantity;
6     double price;
7
8     public Product(int productId, String productName, int quantity, double price) {
9         this.productId = productId;
10        this.productName = productName;
11        this.quantity = quantity;
12        this.price = price;
13    }
14
15    @Override
16    public String toString() {
17        return "ID: " + productId +
18            ", Name: " + productName +
19            ", Quantity: " + quantity +
20            ", Price: " + price;
21    }
22
23 }
```

```
1 package ques1InventoryManagementSystem;
2 import java.util.HashMap;
3 public class InventoryManager {
4
5     HashMap<Integer, Product> inventory = new HashMap<>();
6
7     // Add Product
8     public void addProduct(Product product) {
9         inventory.put(product.productId, product);
10        System.out.println("Product Added.");
11    }
12
13    // Update Product
14    public void updateProduct(int id, int quantity, double price) {
15
16        Product product = inventory.get(id);
17
18        if (product != null) {
19            product.quantity = quantity;
20            product.price = price;
21            System.out.println("Product Updated.");
22        } else {
23            System.out.println("Product Not Found.");
24        }
25    }
26
27    // Delete Product
28    public void deleteProduct(int id) {
29
30        if (inventory.remove(id) != null) {
31            System.out.println("Product Deleted.");
32        } else {
33            System.out.println("Product Not Found.");
34        }
35    }
36 }
```

```
5  
6  
7 // Display Products  
8 public void displayProducts() {  
9     for (Product product : inventory.values()) {  
10         System.out.println(product);  
11     }  
12 }  
13 }  
14 }
```

```
1 package ques1InventoryManagementSystem;
2
3 public class InventoryTest {
4
5     public static void main(String[] args) {
6
7         InventoryManager manager = new InventoryManager();
8
9         manager.addProduct(new Product(101, "Laptop", 10, 55000));
10        manager.addProduct(new Product(102, "Mouse", 50, 700));
11
12        System.out.println("\nInventory:");
13        manager.displayProducts();
14
15        manager.updateProduct(101, 15, 53000);
16
17        System.out.println("\nAfter Update:");
18        manager.displayProducts();
19
20        manager.deleteProduct(102);
21
22        System.out.println("\nAfter Delete:");
23        manager.displayProducts();
24    }
25
26 }
27
```

Product Added.

Product Added.

Inventory:

ID: 101, Name: Laptop, Quantity: 10, Price: 55000.0

ID: 102, Name: Mouse, Quantity: 50, Price: 700.0

Product Updated.

After Update:

ID: 101, Name: Laptop, Quantity: 15, Price: 53000.0

ID: 102, Name: Mouse, Quantity: 50, Price: 700.0

Product Deleted.

After Delete:

ID: 101, Name: Laptop, Quantity: 15, Price: 53000.0